

Pranshu Nijhawan

Principal Software Architect

mail@pranshunijhawan.dev · +91-9999966272 · pranshunijhawan.dev · linkedin.com/in/pranshunijhawan

Principal Software Architect with 10+ years designing scalable **SaaS** platforms and **cloud-native** products at McKinsey & Company, Eptura, and Gartner. Architected data-intensive systems processing 200M+ records across 100+ tables, designed **agentic AI architectures** with tool-augmented **LLM** workflows, and shipped 5 enterprise **SaaS** products powered by advanced analytics, **IoT**, and **machine learning**.

- Core stack: **TypeScript**, **C#**, **Python**, **Spark**, **Databricks**, **PostgreSQL**, **React**, **Node.js**, **Azure**, **Kubernetes**.

Professional Experience

McKinsey & Company

March 2022 – Present

Principal Architect I

Previously: Senior Software Engineer II, Senior Software Engineer I

Principal Architect for Promotion Advisor, a flagship Periscope product serving CPG and retail enterprise clients.

- Agentic AI Architecture:** Designed the **agentic architecture** for Promotion Advisor 2.0 — **Genesis**-based autonomous agents with **MCP (Model Context Protocol)** connectors, **Knowledge Base** grounding, and a **Deep Thinking** execution loop (Observe > Think > Tool Call). Externalized all business logic into configurable **Instruction Sets**, eliminating custom code deployments for client onboarding, and enforced data access control via **Casbin** policies against read replicas.
- Multi-Language Backend Architecture:** Defined the architectural roadmap for a data-intensive backend ecosystem spanning **Node.js/Fastify** (scalable microservices) and **.NET/C#** (high-performance batch processing), enabling independent service deployments across 10+ microservices serving enterprise CPG and retail clients.
- Blue-Green ETL Platform:** Architected a blue-green **ETL** platform (**Databricks**, **Azure Data Factory**, **PostgreSQL**) enabling zero-downtime releases — one production **slot** serves live traffic while the other receives updates. Processes 200M+ records across 100+ tables in under 15 minutes.
- NL-to-SQL AI Prototyping:** Prototyped a natural-language-to-SQL interface for the promotion platform by training a custom encoder-decoder **transformer** (BPE tokenizer, **PyTorch**) and fine-tuning **T5** on a synthetic **NL-to-SQL** dataset generated from production **PostgreSQL** schemas, establishing the technical foundation for AI-powered data querying in the agentic platform roadmap.
- GitOps, Security & Observability:** Established **GitOps**-driven delivery (**Terraform**, **Azure DevOps**, **Helm**, **ArgoCD**) with one-touch release pipelines across 10+ microservices, embedded **security** scanning (**Wiz**, **SonarQube**) into every **CI/CD** workflow, and unified **observability** with **Dynatrace** and **OpenTelemetry** for end-to-end distributed tracing.
- Career Progression & Engineering Mentorship:** Promoted from Senior Software Engineer to **Principal Architect** in 3.5 years. Mentored senior engineers on **system design** trade-offs and drove alignment between product strategy and technical delivery across cross-functional stakeholders.

Technologies: TypeScript · Node.JS · C# · .NET · React · Azure · Databricks · PySpark · SparkSQL · PostgreSQL · Kubernetes

Eptura (formerly Condeco)

April 2021 – March 2022

Principal Engineer

Led the rebuild of Condeco Connect (meeting room booking) from legacy single-tenant to **cloud-native** microservices architecture.

- Cloud-Native Transformation:** Led re-architecture of legacy single-tenant **SaaS** to **cloud-native** microservices on **Azure Kubernetes Service (AKS)**, cutting incident recovery time by 60% and eliminating single-tenant scaling bottlenecks for enterprise client operations.
- Global IoT Release Orchestration:** Engineered and deployed a release orchestration system (**Azure IoT Hub**, **Serverless Functions**, **Azure CosmosDB**) synchronizing **firmware** updates across 300M+ **IoT** devices globally with zero-downtime rollouts.
- Engineering Team Leadership:** Directed a team of 20 engineers (4 tech leads) across four product squads. Established structured **mentorship** program, improving team retention by 20% and accelerating engineering growth across the organization.

Technologies: C# · .NET · React · Angular · Azure · Azure CosmosDB · Azure Functions · SQL Server

Delivered full-stack applications for e-commerce and HR enterprise clients while establishing modern engineering practices across project teams.

- Full-Stack Application Delivery:** Shipped production web applications (**Angular**, **.NET Core**, **PostgreSQL**, **Ionic Framework**) for e-commerce and HR enterprise clients, delivering responsive cross-platform experiences across web and mobile with 99.5%+ uptime.
- CI/CD & Infrastructure as Code:** Established **CI/CD** pipelines and **Infrastructure as Code (IaC)** practices, reducing deployment failures by 70% and ensuring stable, predictable software releases for client projects.

Technologies: C# · .NET · Angular · Ionic Framework · AWS · PostgreSQL · DynamoDB · Fargate

Eptura (formerly Condeco)
Software Engineer

September 2018 – December 2019

Core development and optimization of Condeco Connect (meeting room booking **SaaS**), scaling the platform for a rapidly growing customer base.

- Microservices for 3x Customer Growth:** Designed and developed microservices (**C#**, **.NET**, **React**, **SQL Server**) for the workspace **SaaS** platform, supporting 3x customer scale-up while maintaining system reliability under increased load.
- CI/CD & Data Model Optimization:** Optimized **CI/CD** pipelines reducing build times by 40% and refactored core data models, cutting average query latency by 3x for client-facing features.

Technologies: C# · .NET · React · Angular · Azure · Azure CosmosDB · Azure Functions · SQL Server

Gartner
Associate Software Engineer

February 2016 – August 2018

Previously: Intern

Backend development for the ClearForce platform (proprietary survey and analytics tool for HR consulting), focused on data accuracy and workflow automation.

- SaaS Analytics Platform Development:** Developed backend features for ClearForce (**C#**, **.NET Core**, **React**), improving employee survey data accuracy by 60% and enabling data-driven insights for HR consulting clients.
- SQL Workflow Automation:** Designed and automated SQL workflows (stored procedures, SQL Agent jobs), saving 240+ engineering hours annually in manual data processing and improving data delivery efficiency.

Technologies: C# · .NET · React · SQL Server · AWS

Technical Skills

SaaS & Architecture	System Design, Microservices, Agentic AI, MCP (Model Context Protocol), Data Lakehouse, Event-Driven Architecture, Serverless, Domain-Driven Design (DDD), API Design.
Cloud & Infrastructure	Azure, AWS, Kubernetes, DevOps (Terraform, ArgoCD, Helm), CI/CD, Infrastructure as Code (IaC), Observability (Dynatrace, OpenTelemetry).
Data Engineering & Intelligence	Spark, Databricks, ELT/ETL, PostgreSQL, Real-time Analytics, NL-to-SQL, Text-to-SQL (T5, PyTorch), Data Modeling.
Tech Stack	C#, Python, TypeScript, JavaScript, Node.js, .NET Core, React, Angular, SQL.
Compliance & Security	SOC2, InfoSec, Secure SDLC, Audit Automation, SLA/SLO Management.

Certifications

- Microsoft Certified: Azure Data Engineer Associate**
- Microsoft Certified: Azure Developer Associate**

Education

Bachelor of Technology (B. Tech), Computer Science & Engineering

Gurgaon Institute of Technology & Management · 2012 – 2016